

Tinon Turja Majumder

[GitHub](#) | [Email](#) | [LinkedIn](#) | [Website](#)

Education

Master of Science in Computer Science and Engineering - United International University

January 2024 - December 2025 | **CGPA: 3.61/4.00**

- Relevant Coursework:** Object Oriented Programming, Data Structures and Algorithms, Artificial Intelligence, Big Data, Deep Learning, Machine Learning, Research Methodology, NLP

Bachelor of Science in Textile Engineering (Dyes and Chemicals) - Bangladesh University of Textiles

January 2018 – December 2023

- Relevant Coursework:** Probability and Statistics, Linear Algebra, Integral Calculus, Differential Calculus, Computer Programming

Technical Focus Areas

Applied Computer Vision | AI-Driven Industrial Systems | Resource-Aware AI | Domain-Specific LLMs and RAG

Projects

Smart Dyeing: AI-Driven Decision System for Sustainable Textile Manufacturing (Ongoing)

Funding: Asian Development Bank (ADB) – BIRDI Program

Role: Lead Contributor, ML Modeling and AI Integration

- Designing AI system to reduce industrial dyeing water use by 30–40%
- Building data pipelines from MES logs, sensors, lab, and QC records
- Developing prediction and control models using GBDT and neural networks

Fungal Spore Detection and Classification from Microscopic Images

Tools: Python, PyTorch, Computer Vision, CNNs

- Curated 22-class real-world microscopic dataset
- Built imbalance-aware training pipeline using ResNet-50
- Achieved 89% accuracy on highly imbalanced data

Domain-Specific LLM Fine-Tuning and Retrieval-Augmented Generation

Tools: PyTorch, Hugging Face, QLoRA, FAISS, LangChain, FastAPI, Docker

- Fine-tuned **LLaMA-3-8B** on a medical domain dataset using **QLoRA**, reducing memory usage by ~70%
- Implemented a **RAG pipeline** using FAISS to improve factual grounding and reduce hallucinations
- Designed evaluation workflows to compare baseline and retrieval-augmented responses

Work Experience

Intern — Fakir Knitwears Ltd.

March 2023 - May 2023

- Observed industrial dyeing, QC, and large-scale production systems; experience later motivated Smart Dyeing project.

Part-time Lecturer — Udvash Academic and Admission Care

March 2018 - June 2024

- Taught math and analytical reasoning to 10,000+ students across online and offline programs.

(Concurrent with Studies)

Publications (Preparing for Submission)

• An Intelligent Irrigation Decision Support System using IoT and Weather Data [LaTeX]	Supervised by: Dr. Md. Motaharul Islam
• Automated Density-Based Splitting of Merged Clusters [LaTeX]	Supervised by: Dr. Mohammad Nurul Huda
• Fabric Defect Detection Using Histogram Equalization and Convolutional Neural Network [LaTeX]	Supervised by: Dr. Mohammad Nurul Huda
• Learned Query Optimization in Modern Database Systems: A Comprehensive Survey [Paper]	Supervised by: Dr. Mohammad Rezwanul Haque

Technical Skills

Data Science & ML:	Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Machine Learning, Statistical Analysis, Deep Learning, Model Evaluation
NLP & LLMs:	Language modeling, tokenization, word embeddings, transformer-based models, Hugging Face Transformers, retrieval-augmented generation (RAG)
Computer Vision:	Convolutional neural networks (CNNs), image classification, object detection (YOLO), dataset curation and labeling
Data Systems & Deployment:	SQL, relational databases, vector databases (FAISS), API development (Flask, FastAPI), Docker, GitHub version control
Frameworks & Libraries:	PyTorch, TensorFlow, LangChain, Hugging Face, OpenCV

Standardized Test Scores

GRE: 313/340	September 9, 2024
Quant: 162, Verbal: 151, AWA: 3.5	
IELTS: 7.5/9.0	November 15, 2024
Listening: 8.5, Reading: 7.5, Speaking: 7.0, Writing: 7.0	

Certifications

Machine Learning Specialization Certificate by DeepLearning.AI and Stanford University	January 6, 2025
IBM Generative AI Engineering Professional Certificate	January 17, 2026

Extra-Curricular Activities and Affiliation

Vice President, Ekattor Cultural Club, BUTEX	January 2022 - March 2023
Deputy Chief, Logistics, BUTEX Business Club	June 2020 - February 2022
Joint Secretary, BUTEX Career Club	October 2021 - June 2022